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MSDSE: Predicting drug-side effects based on multi-scale features and deep multi-structure neural network

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ABSTRACT

Unexpected side effects may accompany the research stage and post-marketing of drugs. These accidents lead to drug development failure and even endanger patients' health. Thus, it is essential to recognize the unknown drug-side effects. Most existing methods in silico find the answer from the association network or similarity network of drugs while ignoring the drug-intrinsic attributes. The limitation is that they can only handle drugs in the maturation stage.

To be suitable for early drug-side effect screening, we conceive a multi-structural deep learning framework, MSDSE, which synthetically considers the multi-scale features derived from the drug. MSDSE can jointly learn SMILES sequence-based word embedding, substructure-based molecular fingerprint, and chemical structure-based graph embedding. In the preprocessing stage of MSDSE, we project all features to the abstract space with the same dimension. MSDSE builds a bi-level channel strategy, including a convolutional neural network module with an Inception structure and a multi-head Self-Attention module, to learn and integrate multi-modal features from local to global perspectives. Finally, MSDSE regards the prediction of drug-side effects as pair-wise learning and outputs the pair-wise probability of drug-side effects through the inner product operation. MSDSE is evaluated and analyzed on benchmark datasets and performs optimally compared to other baseline models. We also set up the ablation study to explain the rationality of the feature approach and model structure. Moreover, we select model partial prediction results for the case study to reveal actual capability. The original data are available at *http://github.com/yuliyi/MSDSE*.

1. Introduction

When drugs are taken in normal doses, side effects unrelated to the drug's purpose usually occur, causing harm to patients. It is even more unacceptable for a drug to be withdrawn after approval due to unanticipated side effects [1]. Throughout the drug development cycle, determining the side effects of candidates is a crucial factor for success. Therefore, detecting potential drug-side effects early and with precision can enhance safety and mitigate research and development risks [2]. Traditional laboratory methods rely on extensive pharmacological tests to discover side effects, and the validation process is time-consuming and costly [3].

In the past few years, accompanying the abundance of biomedical information, computer-aided methods have been widely used in drug discovery. These methods utilize multi-source drug features and algorithmic models for efficient and accurate target prediction. Clearly, building computer models to identify the possible drug-side effects can make up for the lack of trials. Methods for identifying drug-side products can be summarized from two aspects: features and algorithms.

Commonly used drug feature mainly includes information about the drug-intrinsic (e.g., drug chemical substructure, drug molecular descriptor) and drug-associated entity information (e.g., drug target protein, drug indication, drug pathway) [4]. Bender et al. [5] made the first attempt to predict hundreds of classes of side effects from drug chemical structures alone and demonstrated the feasibility of using drug chemical structures to predict side effects. Huang et al. [6] exploited drug-target interactions, gene ontology, and protein-protein interactions to construct a model using Logistic Regression (LR) and Support Vector Machine (SVM). Liu et al. [7] combined drug-side effects, drug indications, drug pathways, chemical structures, and target proteins to predict adverse drug reactions. Iwata et al. [8] analyzed the correlation between protein structural domains and side effects using the interaction network between drugs and targets. Yamanishi et al. [9] combined chemical structures and drug target proteins into a unified framework

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for side effect prediction. Experiments showed that integrating multiple features significantly improved prediction performance. Huang et al. [10] combined drug substructures and protein-protein interaction networks to build model using the support vector machine. Xiang et al. [11] built a naive Bayesian predictor employing the side effect-gene association that involves side effects, approved drugs, and genes. Zheng et al. [12] integrated drug-interacting proteins, drug-associated genes, and chemical structures to identify potential drug-side effects. Zheng et al. [13] fused the biological space for drug-target interactions, other spaces for drug substituents, and the chemical space for molecular structure into a uniform integrated model to generate the characteristics of drugs from different perspectives. Activity in metabolic networks [14,15] has also been shown to facilitate side effect prediction. Overall, the biological network composed of drugs and associated entities can significantly enhance the predictive accuracy. However, these methods require that drugs possess rich prior knowledge. For the side effect, representation is relatively unitary. Side effects are generally expressed as terms or text [16,17]. Some methods [18–20] treat side effects as labels and describe the task of identifying potential side effects as a multi-label problem.

Computer methods rely primarily on machine learning and deep learning algorithms. Methods based on traditional machine learning generally use multiple drug features or construct correlations and similarities between multi-feature information. Pauwels et al. [21] used chemical structure features for four models, namely support vector machine (SVM), nearest neighbor (NN), sparse canonical correlation analysis (SCCA), and ordinary canonical correlation analysis (OCCA). Liu et al. [7] integrated various drug-associated information and constructed five predictors by using machine learning algorithms: naïve Bayes (NB), logistic regression (LR), K-nearest neighbor (KNN), naïve Bayes (NB), SVM, and random forest (RF). Bresso et al. [22] utilized inductive logic programming to explore side effects common to a significant number of drugs based on integrated drug and target descriptors. These methods consider the problem only from the drug perspective. In contrast, the recommendation system regards the prediction of drug-side effects as pair-wise learning, with drug-side effect pairs serving as individual samples. Cheng et al. [23] constructed a bipartite graph consisting of drug side effect association networks and devised a computational method, phenotypic network inference model (PNIM), to predict potential side effects of drugs by network-based resource allocation process. Zhang et al. [24] built a algorithmic model called feature-derived graph regularized matrix factorization (FGRMF) to reveal the latent associations of drug-side effects. Ding et al. [25] constructed multiple kernels derived from drug-side effect space and fused by Kronecker Regularized Least Squares (Kronecker RLS) to further identify associations of drug-side effects. In addition, drug similarity can be calculated by the drug associated entities. Zhang et al. [26] explored the linear neighbor relationship to compute drug similarity and transferred the similarity to form a similarity-based graph. Zheng et al. [12] proposed to integrate multiple drug information using the bipartite local model (BLM) to measure the drug-drug similarity. Zheng et al. [13] presented a new approach for sampling highly reliable negatives from the unlabeled drug set based on the comprehensive drug similarity. Mashcool et al. [27] implemented a network and domain based recommender system algorithm (NDDSA) for predicting the possible side effects of candidates. Although these methods have tried various combinations of features and algorithms, the performance is still unsatisfactory. Especially with the continuous expansion of biological big network data, the applications based on machine learning seem powerless.

Deep learning technology was widely employed in the field of drug side effect discovery and offer more effective solutions. Zhang et al. [16] applied a restricted Boltzmann machine-based method (RBMBM) and an integrated neighborhood-based method (INBM), and converted the prediction into a recommender system task. Sanjoy et al. [28] constructed a deep learning model that simultaneously recognizes ADRs and related molecular substructures without the prior definition of substructures. Zhao et al. [29] developed a deep neural network named SDPred that combined multiple similarities and convolutional neural network for determining side effect frequencies for drugs. Likewise [30], proposed a novel graph attention model for predicting frequencies of potential side effects. Qian et al. [31] utilized the drug side effect space to construct the Restricted Boltzmann machine (RBM) and integrated the predictions from RBMs by adopting the average decision rule. Yu et al. [32] developed a hybrid graph neural network to integrated substructure sequence, chemical structure, and topology information from drug-side effect network. Xuan et al. [33] exploited various similarities and associations related to drug-side effects to construct heterogeneous graphs, and built a model called GCRS based on graph convolutional autoencoder (GCA) to identify associations of drug-side effects.

However, the performance of these approaches based on similarity (e.g., drug-drug similarity), association (e.g., drug-side effect association, drug-disease association), and interaction (e.g., drug-target protein interaction, drug-drug interaction) rely heavily on the quality of data on drug-associated biological entities. This leads them to be limited to discovering unknown side effects of "familiar" drugs. Nevertheless, in the initial phase of drug discovery, prior knowledge of candidates is quite scarce, and researchers can only obtain chemical structure attributes to build the prediction model. Consequently, we conceive a novel deep learning framework that is unrestricted by prior knowledge and can be applied to the early screening stage of drugs. In this study, a novel predictor, MSDSE, is initiated for learning and merging multi-scale drug-intrinsic features. The contributions of the model include the following:

- Three approaches are applied to extract the diverse intrinsic features from SMILES sequence, chemical substructure, and molecular structure, respectively. Among them, the drug substructure identifiers are coded word embedding vectors, the statistics of the drug substructure are represented as molecular fingerprints, and the drug chemical structure is graph-structured data, which can obtain chemical information by adopting various graph embedding neural network models.
- We construct the convolutional neural network module with the Inception structure to encode the local information of multi-scale features. Inception is designed into four branches stacked by the basic convolution operation unit to form neural networks with different depths, which are leveraged to abstract deeper features from different ranges of receptive fields.
- Taking into account that the aforementioned multiple drug representations contribute diversely to predicting associations of drug-side effects, we set up a multi-head Self-Attention module to discriminate feature contributions and fuse them to produce global features.
- We deem drug-side effect prediction as pair-wise learning, which means that the task objectives are drug-side effect pairs. For the drug features, we dimensionally align the features obtained by the above two processing methods and add them bit by bit. The side effect features adopt unique representation vectors. Based on the representations of both parties, we employ the inner product operation to yield the drug-side effect pair-wise probabilities. The experimental section illustrates that MSDSE outperforms the state-of-the-art methods. The final ablation study and case study prove the robustness and effectiveness of the model.

2. Method

2.1. Problem formulation

The problem is given the premise of M known side effects to determine what potential side effects N drugs will produce. The specific definition given in computer language is to determine the link $(\mathrm{D}, \mathrm{S}, \mathrm{Y}) = \{(d_i, s_j, y_{ij}) \big| d_i \in D, s_j \in S, y_{ij} = 1 \ if \ \exists (d_i, s_j), y_{ij} = 0 \ if \ \nexists (d_i, s_j)\}$

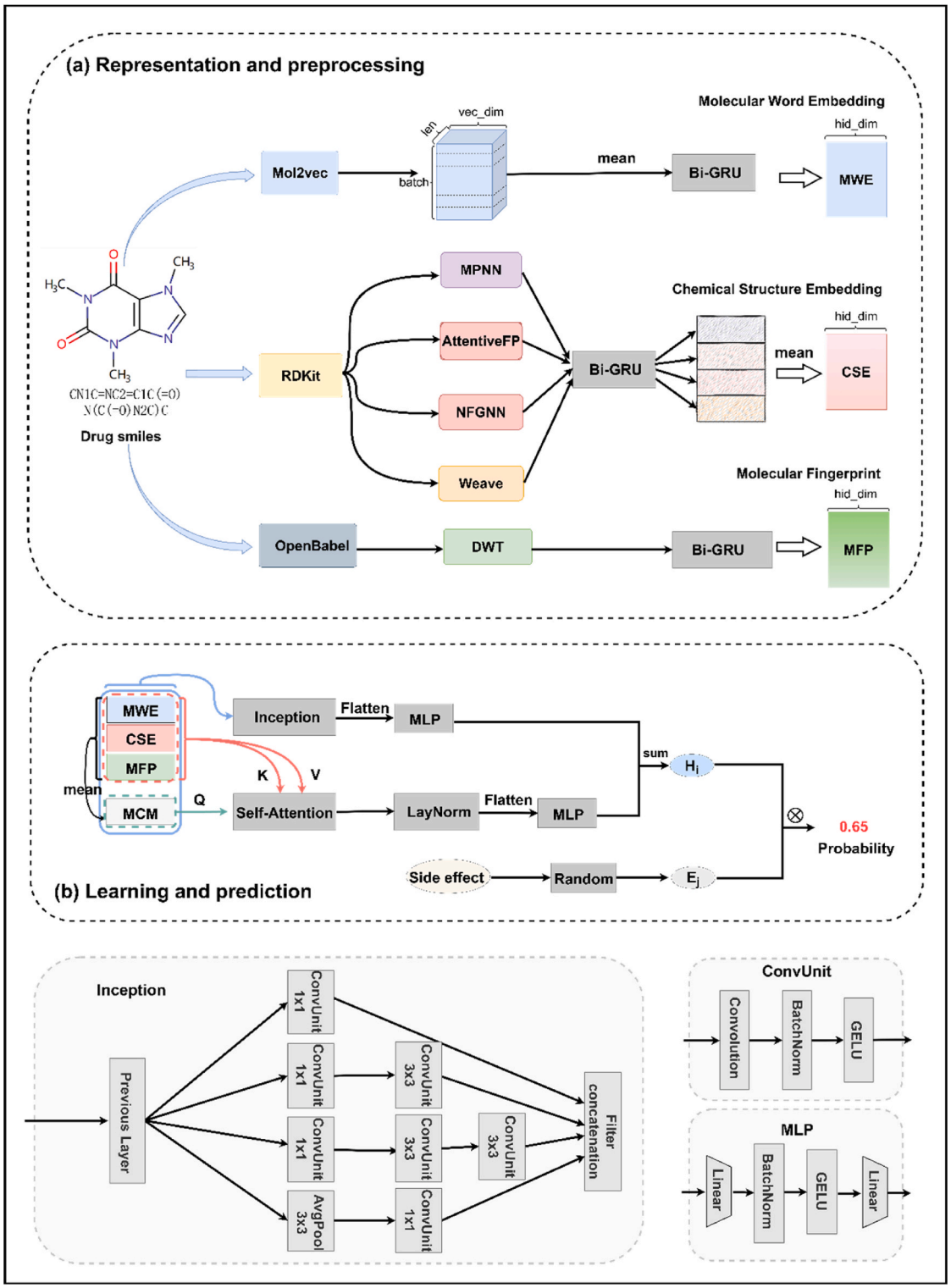


Fig. 1. The entire framework of MSDSE. (a) Representation and preprocessing. MSDSE employs three feature methods to extract rich original representations from the multi-attribute perspective of drugs. They are Mol2vec to convert SMILES string into the word embedding *MWE*, four graph embedding algorithms (i.e., MPNN, AttentiveFP, NFGNN, Weave) to extract chemical structure embedding *CSE* from molecular graph acquired by the RDKit tools, and chemoinformatics algorithm Open Babel to calculate molecular fingerprint representation *MFP* of drug substructures. Subsequently, MSDSE sets up three Bi-GRUs to preprocess the original features respectively. (b) Learning and prediction. We design a bi-level strategy architecture that jointly learns and fuses multi-scale features. The upper part stacks three features and their mean feature *MCM* and inputs them to the Inception module, which is a multi-receptive field convolutional neural network assembled from multi-branch convolution units with multi-size convolution kernels. In the lower part, MSDSE feeds the concatenation of the three features as *K* and *V*, and the mean feature *MCM* as *Q*, into the multi-head Self-Attention module. The bi-level features are added element-wise as the drug feature and the inner product with the side effect feature for the drug-side effect prediction.

between drug node set $D = \{d_i|1 \le i \le N\}$ and side effect node set $S = \{s_j|1 \le j \le M\}$ in the bipartite graph B. For representation, the drug is attached with SMILES representing chemical formula that reflects the sequence structure and graph structure of the molecule, and the side effect is text-type phrase from the MedDRA dictionary. Ultimately, MSDSE learns a mapping $f : (\mathrm{D}, \mathrm{S}) \rightarrow \mathrm{Y}$ of pairs (D, S) and states Y to output connection probability between drug d_i and side effect s_j.

2.2. Overview of MSDSE

The framework structure is depicted in Fig. 1. MSDSE comprises two parts: primary representation and preprocessing stage, feature learning and object discrimination stage. The rudimentary representation of the drug is derived from three calculation approaches. The molecular fingerprinting counts the statistics of drug substructure. The word embedding encodes drug substructure descriptors. The graph embedding learns drug molecular structure graphs. The side effect is denoted by the standardized stochastic vector. Subsequent preprocessing, we unify the dimensions of the original features. The feature learning module applies a bi-level neural network structure to transform the drug representation into a more abstract feature. MSDSE assembles a convolutional neural network with the Inception structure focusing on local features and a multi-head Self-Attention neural network capturing global features to merge the three drug representations. Finally, MSDSE splices drug eigenvectors and side effect eigenvectors to stand for drug-side effect pairs and feeds them into a multi-layer linear neural network to output the association probability. Next, we elaborate on MSDSE.

2.3. Primary representation module

2.3.1. Molecular word embedding

The SMILES is a format using short ASCII strings to represent the structure of chemical species [34]. Atomic identifiers are sorted into "molecular sentences" in the order as in the canonical SMILES. Just as words with similar semantics have similar representations assumed by natural language, Mol2vec [35] treats compounds as sentences and substructures as words, and postulates that similar compound substructures will get similar vector representations. Mol2vec adopts word2vec [36] to perform unsupervised pre-training on the compound corpus composed of ZINC [37] and ChEMBL [38], and derives the vector lookup table of substructures. We apply skip-gram, one of the training modes of word2vec, to encode the Morgan [39] identifiers in the compound sequences to generate substructure vectors.

Before training, each of the v identifiers is encoded into a one-hot vector $z \in \mathbb{R}^{1\times v}$. In the context window of length L, the correlation probability distribution of the central word and other words is calculated by the central word matrix $W_1 \in \mathbb{R}^{v\times dim}$ and the context word matrix $W_2 \in \mathbb{R}^{dim\times v}$. Set the target loss function ℓ_m so that the correlation probability inside the window is as large as possible, while the probability outside the window is as small as possible:

$$z_j^{'} = \left(z_j W_1\right) W_2 \tag{1}$$

$$\ell_m = -\sum_{j=1}^{v} \sum_{k\in \mathrm{C}(j),\ k\notin \mathrm{C}(j)}^{L} \left(log\left(\sigma\left(z_j^{'}(k)\right)\right) - log\left(\sigma\left(z_j^{'}(\widehat{k})\right)\right)\right) \tag{2}$$

where $z_j^{'} \in \mathbb{R}^{1\times v}$ signifies the correlation distribution between the j th central word and the v identifiers, $\mathrm{C}(j)$ represents other identifiers within the context window of the j th identifier, $z_j^{'}(k)$ stands for the correlation between the j th identifier and the k th identifier (positive sample), $z_j^{'}(\widehat{k})$ represents the correlation of the j th identifier with other identifiers (negative samples) outside the context window, σ indicates the *sigmoid* activation function, *dim* is the hidden layer dimension ($dim \ll v$). Finally, the weight W_1 obtained by minimizing ℓ_m is applied to produce the low-dimensional word embedding H_i of drug d_i:

$$H_i = \underbrace{\left[z_j \times W_1, \cdots\right]}_{len(i)} \tag{3}$$

where $len(i)$ denotes the SMILES string length. Owing to the inconsistent length of SMILES, to facilitate the tensor operation of the neural network, we compute the mean value of the molecular word embedding H_i along the string direction to ensure that the embedding of each drug is one-dimensional and equal in length. The molecular word embedding is marked as H_i^{mwe}.

2.3.2. Chemical structure embedding

Drug molecular structure describes the arrangement of atoms and the types of chemical bonds that connect the atoms. Given the SMILES for drug d_i, we utilize the open-source chemical information package RDKit [40] to generate a chemical structure diagram of the drug $g_i = (\nu, \varepsilon)$, where ν represents atom/node set, ε represents chemical bond/edge set. To accurately express the physicochemical properties of drugs, many previous studies have tied rich profiles for ν and ε. For example, MUFFIN [41] adopts the chirality and number of atoms and the direction and type of bonds. SSIDDI [42] only utilizes atomic attributes, including degree, type, formal charge, hydrogens, radical electrons, hybridization, and aromatic. In this work, we refer to the physicochemical properties adopted in idse-HE [32] to describe the molecular attributes of drugs.

To abstract the chemical structure embedding of drug d_i from graph g_i, we employ pre-trained graph embedding algorithms made up of MPNN [43], AttentionFP [44], Weave [45], and NF [46]. The graph embedding process is divided into the message passing and readout phases. The message passing phase formulates an aggregation function M_t and an update function U_t, which are used in turn to integrate the message within the fixed neighbor of the node, and refresh the node's own information. Consequently, the message passing is formulated as follows:

$$m_v^{t+1} = \sum_{w\epsilon N(v)} M_t\left(h_v^t, h_w^t, e_{vw}\right) \tag{4}$$

$$h_v^{t+1} = U_t\left(h_v^t, m_v^{t+1}\right) \tag{5}$$

where t is iteration round, $v \in V$ indicates a node in the graph g_i, $N(v)$ stands for the adjacent nodes of v, h_x^t represents the intermediate state of v at time t, e_{vw} denotes the representations of edge between v and w. Finally, the readout phase applies a readout function R to calculate the eigenvector H_i for graph m_i base on node statuses at t step. The readout phrase is described as follows:

$$H_i^* = R\left(h_v^t | v \in V\right), * \in \{mpnn, weave, afp, nf\} \tag{6}$$

It should be noted that various algorithms have different specific implementations for these two phases. The distinction for message passing lies in the depth, parameter dimension, and hierarchy of the linear layer and the class of the activation function. For the readout phase, MPNN employs Set2Set [47] containing LSTM network that is insensitive to the permutation of nodes to ensure that graph embedding remains invariant to graph isomorphism (aka. sequence invariance). Weave reconstructs each dimension of the node feature into a Gaussian distribution and then sums them up. AFP and NF build multilayers of global pooling modules.

Based on the graph embeddings derived from these graph models, we apply bit-wise mean to output the chemical structure embedding H_i^{cse} for drug d_i:

$$H_i^{cse} = \frac{1}{4}\sum H_i^* \tag{7}$$

2.3.3. Molecular fingerprint representation

The molecular fingerprint [48] is an abstract representation, which

transforms the molecule as a bit string (bit vector) to facilitate comparison across molecules. The prototypical process is to extract the structural features of the molecule and generate a bit vector through Hashing. Each bit on the molecular fingerprint corresponds to a molecular fragment, and the molecular structure is converted into a serial binary fingerprint by checking whether the substructure is present, where 1 indicates that the substructure fragment exists, and 0 indicates no. Open Babel [49] is a chemistry toolbox that implements an extensive library of cheminformatics algorithms. According to the SMILES, OpenBabel adopts a path-based method to calculate the molecular fingerprint of the drug to provide the function of indexing the chemical structure. OpenBabel involves four kinds of fingerprints: FP2, FP3, FP4, and MACCS. In this study, we adopt the FP2 fingerprint format to identify all linear and circular substructures (except for the 1-atom C and N) in molecules of length 1 to 7, and map them to bitstrings of length 1024 using Hashing. To avoid the sparseness of the fingerprint, it is usually converted into a 256-bit hex string s.

In addition, the initial molecular fingerprint features may involve some noise and redundancy that will be detrimental to the final result. To eliminate these negative factors, we utilize the Discrete Wavelet Transform (DWT) [50] to strip the most discriminative feature from the original feature s. In effect, DWT can decompose the signal s into two set coefficients at different dilations. One is the set of approximation coefficients that characterizes the low-frequency and high-scale components of s, i.e., valid information; the other is the set of detail coefficients, i.e., valueless noise. Finally, we concatenate the 128 approximate coefficients as the eigenvector of drug d_i, denoted as H_i^{mfp}.

2.3.4. Side effect vector

To ensure the uniqueness of side effects and simplify the representation method, we use standardized stochastic vectors with mean 0 and standard deviation 0.1 as the side effect representation E_j.

2.4. Drug feature learning module

2.4.1. Preprocessing

In view of that the dimensions of the above original features are inconsistent, we employ bidirectional GRU to project the features into a low-dimensional representation space:

$$\widetilde{H}_i^* = BiGRU^*\left(H_i^*\right), * \in \{mfp, mwe, cse\} \tag{8}$$

After unifying the dimensions of the three different latent vectors, we concatenate them as the hidden feature $\widetilde{H}_i$ for drug d_i, and element-wise average them to generate the other new hidden vector $\overline{H}_i$, and then input both into the downstream feature fusion stage:

$$\widetilde{H}_i = \widetilde{H}_i^{mfp} \bigoplus \widetilde{H}_i^{mwe} \bigoplus \widetilde{H}_i^{cse} \tag{9}$$

$$\overline{H}_i = \frac{1}{3}\sum \widetilde{H}_i^* \tag{10}$$

Next, we conceive two routes to explore the fusion of multi-scale features. One is to use the local receptive field property of the convolution kernel to mine the underlying local information. The other is to leverage the computational mode of the multi-head Self-Attention mechanism to overview the global information.

2.4.2. Local feature learning

Inspired by the GoogLeNet [51] model, we erect an Inception network to fetch the local information of drugs. Inception is a convolutional neural network structure, which sets convolution kernels with several sizes to fulfill multiple receptive fields and finally splicing to fuse features. Since the length 4 for drug feature $\widetilde{H}_i$ and $\overline{H}_i$, we adopt convolution kernels with sizes of 1×1, 3×3 and 5×5. Moreover, multiple low-sized convolution kernels can achieve the same receptive field as large-sized ones. We replace 5×5 with two 3×3 to cut-down parameters and mitigate overfitting (Fig. 1). Among them, each convolution operation is an independent convolution module *ConvUnit*, which includes a single-layer CNN, a batch normalization BN, and a non-linear activation function GELU. A single Inception network structure is shown below:

$$ConvUnit = \{CNN, BN, GELU\} \tag{11}$$

$$C_1 = ConvUnit_{1\times1} \tag{12}$$

$$C_2 = \{ConvUnit_{1\times1}, ConvUnit_{3\times3}\} \tag{13}$$

$$C_3 = \{ConvUnit_{1\times1}, ConvUnit_{3\times3}, ConvUnit_{3\times3}\} \tag{14}$$

$$C_4 = \{AvgPool_{1\times1}, ConvUnit_{1\times1}\} \tag{15}$$

$$Inception = C_1 \bigoplus C_2 \bigoplus C_3 \bigoplus C_4 \tag{16}$$

Finally, we concatenate $\widetilde{H}_i$ and $\overline{H}_i$ as model input to yield local feature H_i^L for drug d_i as the following procedure:

$$H_i^L = Inception(\widetilde{H}_i \bigoplus \overline{H}_i) \tag{17}$$

2.4.3. Global feature learning

MSDSE, with the multi-head Self-Attention mechanism, calculates the feature importance and performs weighted fusion to observe all features from a global perspective. In the Self-Attention module, the length of the output is determined by the input variable Q. Therefore, we take the mean feature vector $\overline{H}_i$ as the input Q, and the concatenated feature matrix $\widetilde{H}_i$ as the input K and V. Evidently, the model outputs a one-dimensional feature vector that fuses all features. Subsequently, MSDSE adds a layer normalization LN to transform the output of the Self-Attention layer and yields the normalized global feature H_i^G of drug d_i:

$$H_i^G = LN(Atten(Q = \overline{H}_i, K = \widetilde{H}_i, V = \widetilde{H}_i)) \tag{18}$$

2.5. Object discrimination module

There is a vast difference in the output dimensions of Inception and Self-Attention. MSDSE sets up multi-layer perceptron *MLP* to map the above two learned information to the same feature space, and then sums the two features to produce the final representation H_i of drug d_i:

$$MLP = GELU\left(BN1d\left(H_i^* W_1\right)\right) W_2, * \in \{L, G\} \tag{19}$$

$$H_i = \frac{1}{2}\left(MLP^L\left(H_i^L\right) + MLP^G\left(H_i^G\right)\right) \tag{20}$$

where H_i^* represents the input tensor, W_1 and W_2 are the weight matrices. *GELU* indicates the leaky version of the Gaussian Error Linear Unit, $BN1d$ denotes one-dimensional batch normalization.

Based on the eigenvector H_i of drug d_i and the eigenvector E_j of side effect s_j, we apply the scalar value of the element-wise inner product to judge the association probability between drug d_i and side effect s_j:

$$y_{ij}^{'} = H_i \times E_j^T \tag{21}$$

3. Results

3.1. Datasets

Drugbank [52] (https://go.drugbank.com/) is a database of drugs and related knowledge, containing approved small molecule drugs, approved biologics, and experimental drugs. SIDER [53] (http://sideeffects.embl.de/) incorporates information on marketed drugs and their recorded adverse drug reactions (ADRs). The current release of

Table 1
The drug-side effect dataset.

Field	Value
Number of drugs	1020
Number of side effects	5599
Number of DSPs	133,750
Average side effects per drug	131.13
Average drugs per side effect	23.89

SIDER 4 comprises more than 1,400 drugs and their reported ADRs. The ADR records contain the PubChem IDs for drugs and the names for side effects taken from MedDRA (https://www.meddra.org/), a clinically validated international medical terminology dictionary. These data come from clinical trials, patient condition monitoring, and electronic health record mining. PubChem (http://pubchem.ncbi.nlm.nih.gov) is an open experimental data repository for validating the biological activity of small molecules, including toxicity and safety information, physical and chemical properties, literature citations, patents, etc. Timilsina et al. [54] constructed a drug-side effect pair dataset by associating the Drugbank database with the SIDER database through the mapping relationship between the PubChem ID and the Drugbank ID of drugs in PubChem. The dataset consists of 1,020 drugs, 5,599 side effects, and 133,750 known drug-side effect associations. The entire sample set consists of 5,710,980 samples from the Cartesian product of the drug set and the side effect set. The known drug-side effects associations are taken as positives and the rest as negatives. Obviously, the sample set is highly skewed. The dataset's statistical information is presented in Table 1 and Fig. 2.

3.2. Experiment settings and evaluation

Our task is to screen out potentially unknown drug-side effect pairs among existing drugs and side effects. For this purpose, we adopt ten-fold cross-validation to train model parameters and evaluate model performance. In addition, we randomly split 5% of the data from the training set as a validation set for verifying the hyperparameters. MSDSE iteratively trains end-to-end in each fold until the experimental metrics on the validation set are not improved within patient epoch set to 50. The performance evaluation results of MSDSE are expressed as *mean* ± *standard deviation* from the ten-fold test set.

In the context of the data set being incredibly unbalanced, we select common sensitive indicators in binary classification problems to evaluate model performance, including area under the precision-recall curve (AUPR), Matthews correlation coefficient (MCC), F1 score, and mean reciprocal rank (MRR), where MRR denotes the mean of predicted ranking reciprocal for positive samples. In order to facilitate the display of MRR and other metrics in the same range, we amplify MRR by a multiple of N/10 (N = 133750).

3.3. Hyperparameter search

To ensure that the model achieves the optimal fit, we employ the grid search to investigate the hyperparameters: L2 penalty (i.e., regularization coefficient of the loss function), learning rate, and hidden

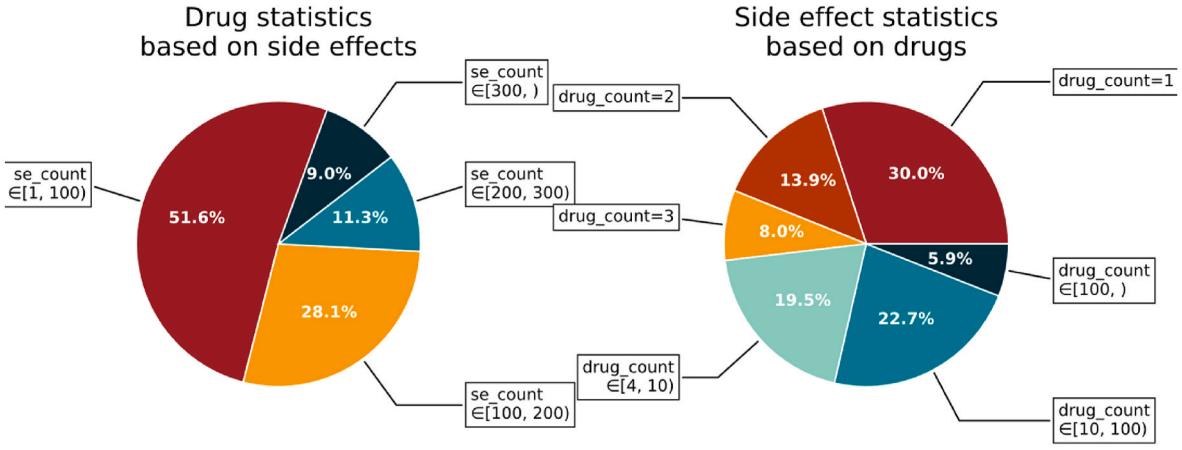


Fig. 2. The statistics of the dataset (The proportion of drugs based on the number range of associated side effects and the proportion of side effects based on the number range of associated drugs).

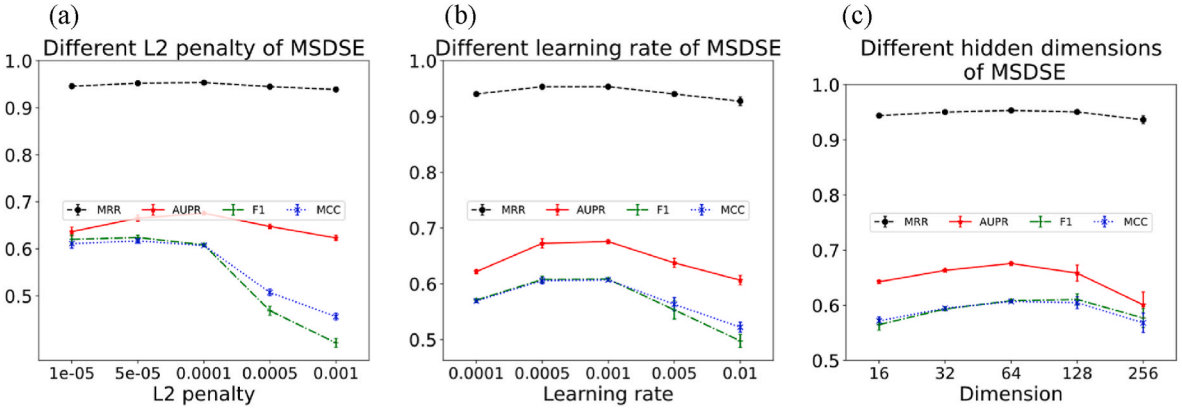


Fig. 3. The performance of MSDSE under different hyperparameters setting. The y-axis represents the predicted value, the x-axis represents each hyperparameter value, and the four polylines with different colors represent the trends of the four evaluation metrics under each hyperparameter value.

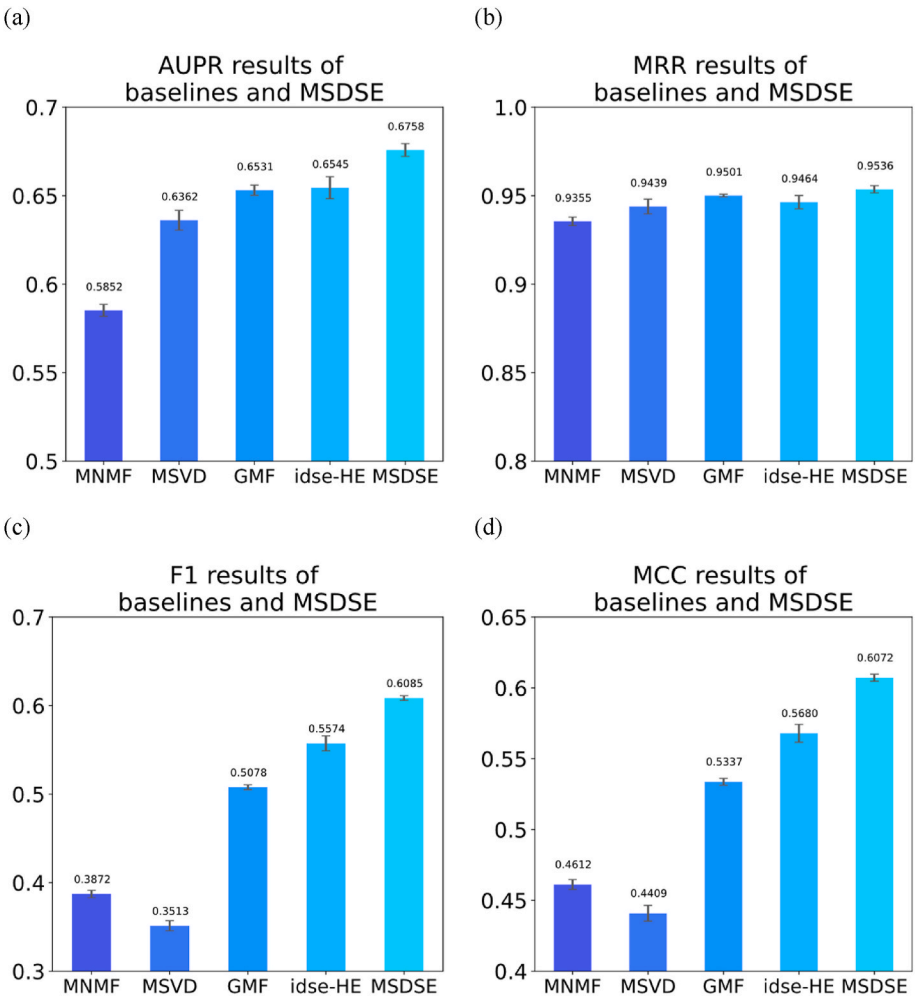


Fig. 4. The results of baselines and MSDSE with four metrics.

dimension. Five equidistant values are chosen within a specific range of hyperparameters. The change trends of various performance indicators of the model under diverse profiles are shown in Fig. 3. Thus, we set L2 penalty to 0.0001, learning rate to 0.001, and hidden dimension to 64.

3.4. Method comparison

To more intuitively comprehend the superior performance of MSDSE, we have chosen the following methods as baselines for comparison:

- MNMF [54]: MNMF reconstructs the drug-side effect adjacency matrix utilizing the Nonnegative Matrix Factorization (NMF) and employs heat diffusion based on the matrix scores to predict the impact of drug-side effects.
- MSVD: MSVD replaces NMF in MNMF with TruncatedSVD.
- GMF [55]: Generalized Matrix Factorization (GMF) is the matrix factorization in the field of deep learning. GMF performs the element-wise multiplication on the potential eigenvectors and feeds them into the linear neural network layer to output the result.
- idse-HE [32]: idse-HE integrates multiple drug features using a hybrid graph neural network and derives predicted values by reconstructing the adjacency matrix of drug-side effects.

The performance of MSDSE and all baselines are displayed in Fig. 4. The statistical results for all metrics show that MSDSE outperforms the runner-up by up to 2.13% on AUPR, 0.35% on MRR, 5.11% on F1, and 3.92% on MCC. The results reflect that the comprehensive performance of our model is optimal. Furthermore, we delve into the performance of the aforementioned methods in all folds. The four metrics of all methods in each fold are shown in Fig. 5. Although MSDSE is slightly inferior to GMF for the fourth fold in the MRR plot, our model exhibits absolute dominance in all folds. This further shows that MSDSE can exert stable performance for predicting various unbalanced data sets.

Essentially, the baselines predict side effects based on the drug-side effect association matrix. MNMF and MSVD used the traditional matrix factorization algorithm, and GMF was a matrix factorization method for deep learning. They all projected the association matrix into the drug and side effect matrices in the lower dimensional space. idse-HE regarded the association matrix of drug-side effects as a bipartite graph, and applied the graph embedding algorithm to extract the embedding of drug and side effect nodes. It can better understand the topological relationship among nodes. More importantly than the first three, idse-HE incorporated the chemical structure information to improve the predictive ability. However, these baselines have two defects: the lack of more profound insight into the drug and the heavy reliance on prior knowledge that is inapplicable for side effect identification of new drugs. As opposed to the associated relationship, our method makes full use of the properties of drugs to adapt to a broader range of application scenarios, including predicting unknown side effects of known drugs and side effects of new drugs. In addition, we construct a multi-structure complex neural network based on the structural characteristics of the feature data to satisfy the learning of original features.

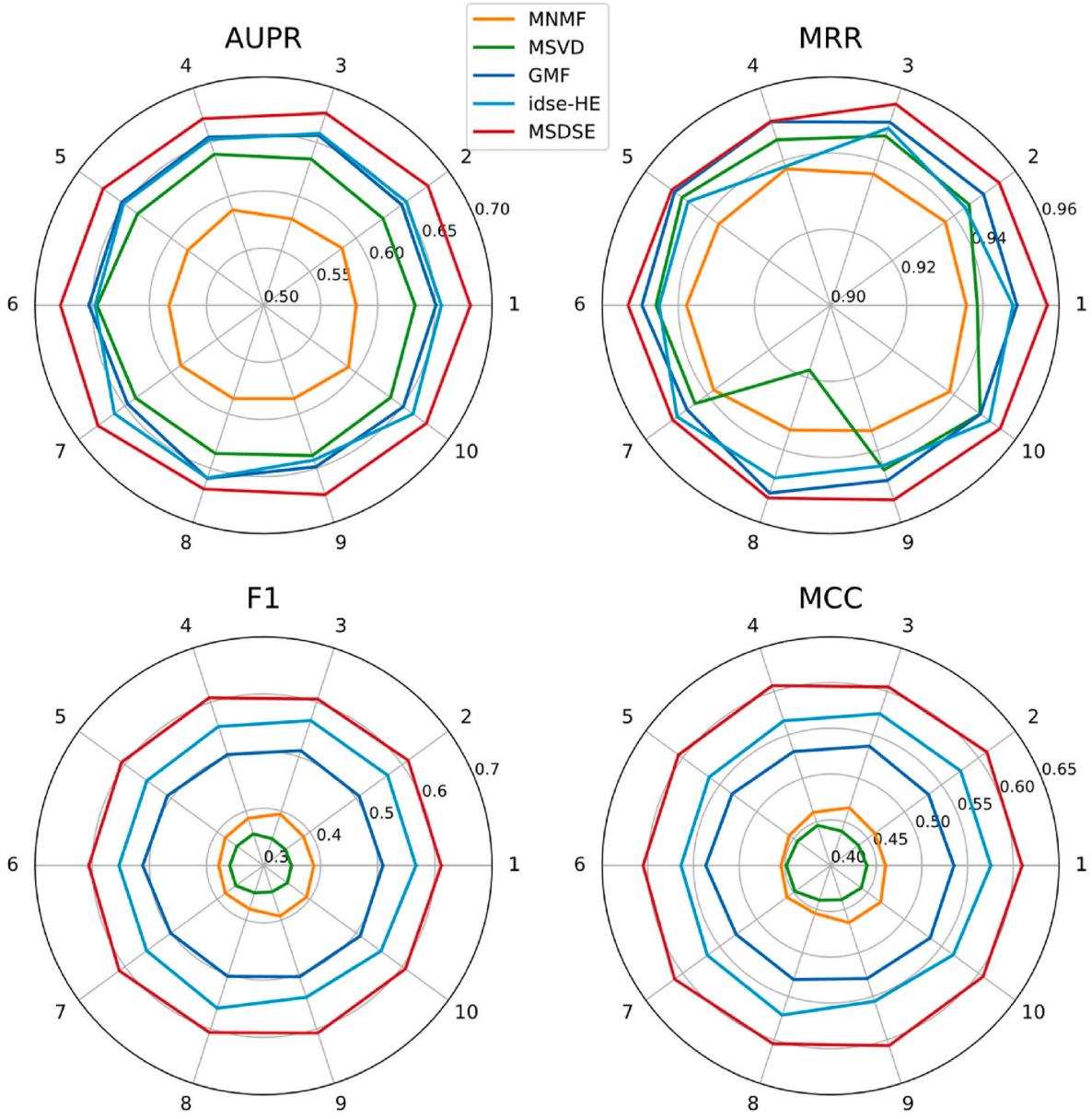


Fig. 5. The ten-fold results of four metrics for baselines and MSDSE. The polar axis (radius) represents the performance of the model on each fold, and the polylines with different colors represent baselines and MSDSE.

3.5. Ablation study

To analyze and demonstrate the contribution of the various modules, we modified the features and structure of the original model, respectively, to obtain seven variants, which are outlined in detail below:

- MSDSE_NO_MFP only removes molecular fingerprints representing drug substructures.
- MSDSE_NO_MWE simply deletes word embedding vectors representing drug substructure "words".
- MSDSE_NO_CSE only eliminates the chemical structure information of drugs extracted by multiple graph embedding algorithms.
- MSDSE_INC removes the Self-Attention module and only relies on the Inception module to process drug features.
- MSDSE_ATT eliminates the Inception module and utilizes the Self-Attention mechanism to integrate multiple drug features.

The predictive performance of MSDSE and the five variants are

Table 2
The performance of MSDSE and all variants.

Method	AUPR	MRR	F1	MCC
MSDSE_NO_MFP	0.6522 ± 0.0159	0.9485 ± 0.0040	0.5795 ± 0.0157	0.5812 ± 0.0141
MSDSE_NO_MWE	0.6653 ± 0.0117	0.9521 ± 0.0027	0.6004 ± 0.0077	0.5992 ± 0.0077
MSDSE_NO_CSE	0.6537 ± 0.0061	0.9459 ± 0.0115	0.5842 ± 0.0043	0.5843 ± 0.0042
MSDSE_INC	0.6573 ± 0.0083	0.9509 ± 0.0020	**0.6126 ± 0.0061**	0.6063 ± 0.0061
MSDSE_ATT	0.6702 ± 0.0109	0.9528 ± 0.0035	0.6039 ± 0.0072	0.6026 ± 0.0071
MSDSE	**0.6758 ± 0.0036**	**0.9536 ± 0.0020**	0.6085 ± 0.0026	**0.6072 ± 0.0025**

The best results are highlighted in boldface.

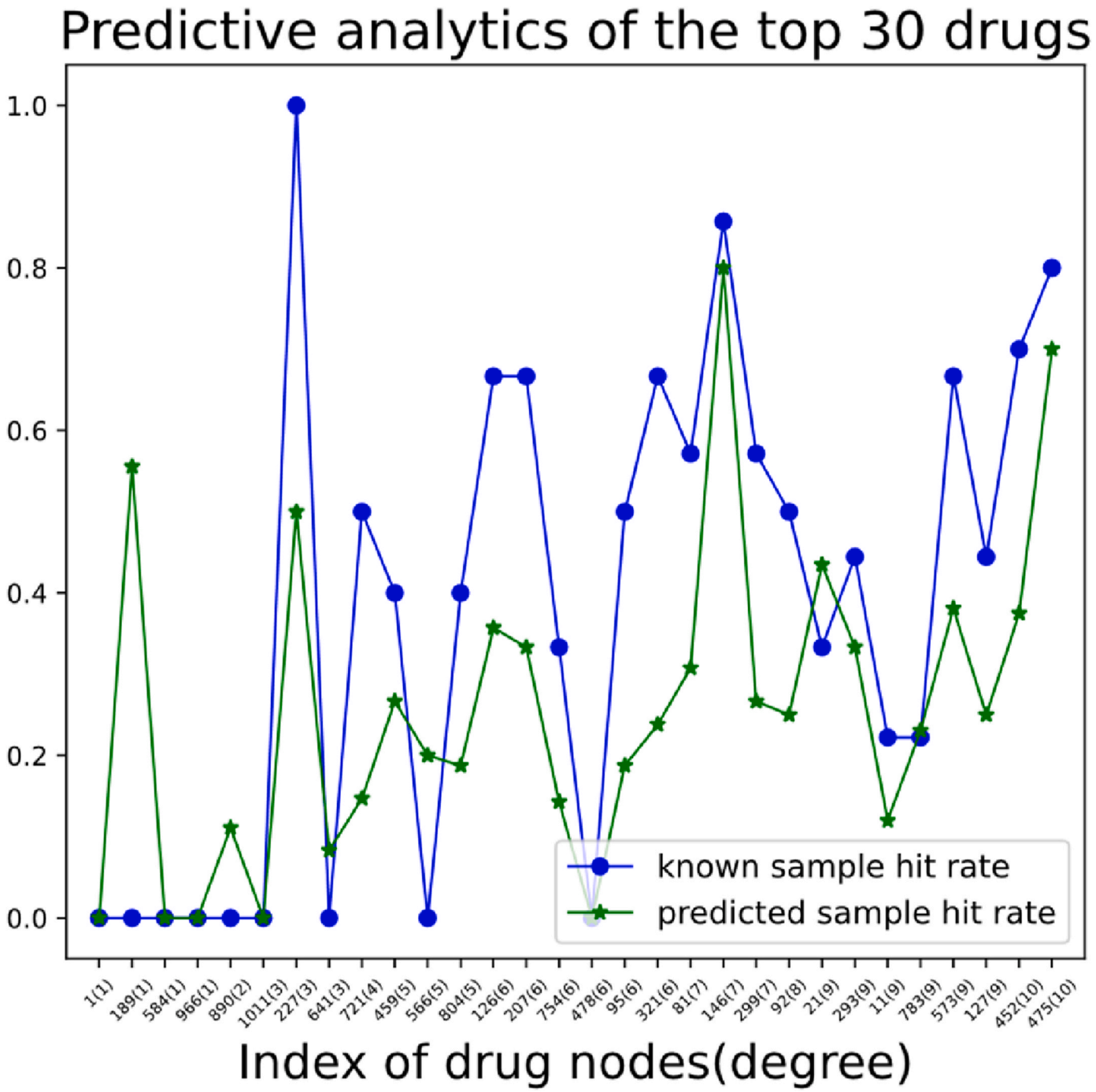


Fig. 6. Predictive analytics of the top 30 drugs in ascending order of the number of caused side effects. The blue line represents the prediction accuracy in known positive samples, and the green line represents the complete accuracy in the predicted positive samples (including known positive samples and confirmed potential positive samples). The x-axis represents the index of nodes (starting at 0) and the degree (i.e., number of associated side effects) of nodes in the bipartite graph *B*.

presented in Table 2. In terms of the feature variants, MSDSE_NO_MWE is optimal, and the value of each metric is relatively close to MSDSE, and the other two variants have declined significantly on F1 and MCC. These show that word embedding contributes minimally, while chemical structure embedding and molecular fingerprint are more crucial. In addition, the fusion of features at different scales promotes the model's performance. As far as the model structure variants, MSDSE_INC has the worst performance on AUPR and reaches the best on F1, and MDSE_ATT is close to MSDSE in all metrics. In conclusion, the MSDSE that integrates the Inception structure and the Self-Attention module can show more desirable performance.

Overall, MSDSE achieves the best state in both features and model structure. By comparing the above variants, we demonstrate that the fusion of diverse features helps to portray drug properties more comprehensively. Moreover, the adaptive neural network is designed according to the data structure of the feature for further processing, and the high-quality features obtained are conducive to subsequent prediction tasks.

3.6. Case study

In the end, we illustrate the actual ability of MSDSE through specific

cases. Specifically, we pay attention to the top 30 of potential drug-side effect pairs, according to descending order of the negative sample scores. We search and verify these predictions in ADReCS [56] database, drugs.com, and literatures. We verify 9 out of the top 10 pairs. Furthermore, 16 out of the top 20 and 24 out of the top 30 are affirmed. Specific drug-side effect pairs and evidence can be viewed in the Supplementary File. For example, we predicted that dyspepsia could be caused when taking pregabalin, which is reported in the ADReCS.

In addition, we observed an extreme scenario where some drugs cause only one or a few side effects. To further explore the performance of the model under such boundary condition, we analyze the top 30 drugs in ascending order of the connection degree of drug nodes. In Fig. 6, both hit rates are almost 0 at the initial stage of low degree for drug nodes. As the known side effects of the drug increase, the two curves shake out of the 0-value region, and the performance of model is significantly enhanced. Overall, the model performs better on known positive samples than on predicted positive samples. The reason is that MSDSE makes more misjudgments in predicting potential side effects.

4. Conclusion

In the study, we propose a multi-structure neural network called MSDSE to predict the potential side effects of drugs. MSDSE adopts three distinct scales of drug information: molecular fingerprints from substructure statistics, word embedding vectors from substructure descriptors, and chemical structure embeddings from molecular graphs. Our model constructs a locally-focused convolution module and a globally-focused Self-Attention module for re-learning features, respectively. The experimental results reveal the superior performance of our model compared to other state-of-the-art models. Furthermore, we demonstrate the correctness of the feature combination and structure design of MSDSE through the ablation study.

Our findings significantly advance in identifying potential side effects for drugs, but only for a single drug. In reality, patients often take several drugs to treat diseases simultaneously. In the following work, we will attempt the side effect prediction in the polypharmacy scenario, and limit it to the drug characteristics to cope with a broader range of drug screening.

Author contributions

Yu and Qiu conceived and formulated the research plan. Yu and Xu drafted and revised the manuscript. Xiao supervised the project. All authors contributed to the content of this paper and approved the final manuscript.

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Code and data availability

The source code and data are provided in the MSDSE GitHub repository, which is *http://github.com/yuliyi/MSDSE*.

Declaration of competing interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2023.107812.

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